Diogo M. Camacho

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Executive Profile

Led and built computational organizations across biotech and pharma for 15+ years, from early-stage venture to scaled platform team. Deep in ML, multi-omics, and computational biology; broad across the discovery stack from target ID to preclinical validation. Drove organization-wide AI/ML adoption, built the computational platform narrative for fundraising, and presented strategy to boards and VC partners. Led global teams spanning the US and Europe. Secured >$2.4M in competitive DARPA and NIH funding; 6 patents; 20+ publications (7,000+ citations).

- **Leadership through ambiguity**: Built and scaled computational organizations across evolving ventures, from an undefined early mandate to a functioning platform team. Comfortable owning technical direction before the path is clear.
- **Technical innovation**: Established AI platforms across the discovery lifecycle: patient-stratification, active-learning frameworks, and agentic systems spanning target ID through preclinical validation.
- **Building and judging talent**: Recruit and develop computational leaders: hiring, mentorship, succession planning, and the executive searches that fill senior gaps. Scaled from first hire to a 25-person group at Cellarity.
- **Executive credibility**: Board and VC interface across multiple companies. Secured >$2.4M in competitive DARPA and NIH funding; 6 patents and 20+ publications (7,000+ citations) across computational biology, ML, and synthetic biology.

Professional Experience

Flagship Pioneering
Cambridge, MA

SENIOR PRINCIPAL, COMPUTATION — *Oct 2024 - Present*

- **Company origination**: AI lead on the founding team of FL111, owning computational efforts from exploration through seed funding. Advisory roles on FL106, FL108, and FL115 across the same stages.
- **Venture platform strategy**: Computational and data strategy for multiple seed-stage ventures. Built platform capabilities and set the technical direction for early programs.
- **Agentic AI systems**: Built agentic AI systems for drug discovery and direct-to-consumer health applications across the Flagship portfolio.
- **Talent and leadership**: Recruited and retained computational science leaders and team members across portfolio companies.

Abiologics *(Flagship Pioneering portfolio company)*
Cambridge, MA

HEAD OF COMPUTATIONAL SCIENCES — *Jan 2025 - Present*

- **Platform strategy**: Lead the computational sciences function for a D-amino acid therapeutic platform: AI-first design strategy, operations, and execution.
- **Active learning**: Made active learning the core experimental-design approach for mini-binder optimization, so candidate prioritization across the DBT cycle runs on data, not intuition.
- **Agentic AI**: Built and deployed agentic AI systems for target identification, prioritization, and competitive intelligence.
- **Organizational leadership**: Own team growth and function end to end: hiring, performance management, budgeting, and executive search.

Cellarity
Boston, MA

VICE PRESIDENT, COMPUTATIONAL AND DATA SCIENCES — *Jul 2022 - Feb 2024*

- **Executive leadership**: On the R&D Leadership Team, reporting to the CSO. Shaped corporate strategy where the computational platform met therapeutic portfolio direction.
- **Organizational build**: Scaled and led a 25-person computational organization across bioinformatics, ML, and computational biology, with teams in Boston and Germany.
- **Platform strategy**: Directed platform strategy and operations end to end. Put Cellarity's computational capabilities into R&D programs and into the investor narrative for fundraising.
- **External engagement**: Primary computational interface for the board, founding VC partners, and external collaborators on platform differentiation and roadmap.

Rheos Medicines
Boston, MA

Senior Director, Computational Biology and Bioinformatics — *Nov 2020 - Jul 2022*

- **Precision medicine strategy**: Set the computational precision-medicine strategy for immunometabolism drug discovery: how patient stratification and multi-omics data would drive program decisions.
- **AI and data strategy**: Defined and implemented Rheos' AI and data strategy with Research Informatics, across discovery and translational programs.
- **Platform delivery**: Built the MetPM platform and MetKG knowledge graph: patient-stratification algorithms and multi-omics tooling adopted across the pipeline.
- **Board engagement**: Presented computational strategy to the board and pharma partners; contributed to the Series B investor narrative.

Wyss Institute @ Harvard University
Boston, MA

Lead, Predictive BioAnalytics — *Jul 2016 - Nov 2020*

- **Founding leadership**: Co-founded and led the Predictive BioAnalytics Initiative, a new ML/AI function embedded in translational science. Set the research strategy, secured the funding, and built the team from scratch.
- **Capital acquisition**: Raised >$2.4M in competitive DARPA and NIH grants as Co-PI.
- **Infectious disease research**: Co-led the computational side of host immune-tolerance research: transcriptomics that identified infection-response pathways, later validated experimentally and published in Advanced Science.
- **Team leadership**: Directed and mentored interdisciplinary teams (staff scientists, postdocs, graduate students, interns) across synthetic biology, systems biology, and ML.

Evelo Biosciences
Cambridge, MA

Senior Scientist, Computational Systems Biology Lead — *Jan 2015 - Apr 2016*

Ember Therapeutics
Cambridge, MA

Principal Scientist, Computational Systems Biology Lead — *Jan 2014 - Dec 2014*

Pfizer, Inc.
Cambridge, MA

Senior Research Scientist, Computational Sciences Center of Emphasis — *Jan 2011 - Jan 2014*

Howard Hughes Medical Institute @ Boston University
Boston, MA

Postdoctoral Fellow (Jim Collins Lab) — *Jul 2007 - Jan 2011*

Education

Virginia Polytechnic Institute and State University
Blacksburg, VA

Ph.D. in Genetics, Bioinformatics, and Computational Biology

Faculdade de Ciencias da Universidade de Lisboa
Lisboa, Portugal

B.Sc. in Biochemistry

Publications and Patents

- Valeri JA, Soenksen LR, Collins KM, Ramesh P, Cai G, Powers R, **Camacho DM**, et al. (2023). BioAutoMATED: An end-to-end automated machine learning tool for explanation and design of biological sequences. *Cell Systems* 14, 525.
- Sperry MM, Novak R, Keshari V, Dinis ALM, Cartwright MJ, **Camacho DM**, Pare JF, Super M, Levin M, Ingber DE (2022). Enhancers of host immune tolerance to bacterial infection discovered using linked computational and experimental approaches. *Advanced Science* 9, 2200222.
- Gazzaniga FS, **Camacho DM**, Wu M, Palazzo MFS, Dinis ALM, Grafton FN, Cartwright MJ, Super M, Kasper DL, Ingber DE (2021). Harnessing colon chip technology to identify commensal bacteria that promote host tolerance to infection. *Frontiers in Cellular and Infection Microbiology* 11, 638014.
- Valeri J, Collins KM, Ramesh P, Alcantar M, Lepe BA, Lu TK, **Camacho DM** (2020). Sequence-to-function deep learning frameworks for engineered riboregulators. *Nature Communications* 11, 5058.
- Bojar D, Powers RK, **Camacho DM**, Collins JJ (2020). Deep-learning resources for studying glycan-mediated host-microbe interactions. *Cell Host & Microbe* 29, 132-144.
- Jalili-Firoozinezhad S, Gazzaniga FS, Calamari EL, **Camacho DM**, Fadel C, Nestor B, Cronce MJ, Tovaglieri A, Levy O, Gregory KE, Breault DT, Cabral JMS, Kasper DL, Novak R, Ingber DE (2019). A complex human gut microbiome cultured in an anaerobic intestine-on-a-chip. *Nature Biomedical Engineering* 3, 520-531.

- Tovaglieri A, Sontheimer-Phelps A, Geirnaert A, Prantil-Baun R, **Camacho DM**, Chou DB, Jalili-Firoozinezhad S, de Wouters T, Kasendra M, Super M, Cartwright M, Richmond CA, Breault DT, Lacroix C, Ingber DE (2019). Species-specific enhancement of enterohemorrhagic E. coli pathogenesis mediated by microbiome metabolites. *Microbiome* 7, 43.
- Bojar D, **Camacho DM**, Collins JJ (2019). Using natural language processing to learn the language of glycans. *Nature Communications* (in review).
- **Camacho DM**, Collins KM, Powers RK, Costello JC, Collins JJ (2018). Next-generation machine learning for biological networks. *Cell* 173, 1581-1592.
- Musah S, Dimitrakakis N, **Camacho DM**, Church GM, Ingber DE (2018). Directed differentiation of human induced pluripotent stem cells into mature kidney podocytes and establishment of a Glomerulus Chip. *Nature Protocols* 13, 1662-1685.
- Paandey SP, Winkler JA, Li H, **Camacho DM**, Collins JJ, Walker GC (2014). Central role for RNase YbeY in Hfq-dependent and Hfq-independent small-RNA regulation in bacteria. *BMC Genomics* 15, 121.
- Galagan JE, Minch K, Peterson M, **Camacho D**, et al. (2013). The Mycobacterium tuberculosis regulatory network and hypoxia. *Nature* 499, 178-183.
- Belenky P, **Camacho D**, Collins JJ (2013). Fungicidal drugs induce a common oxidative-damage cellular death pathway. *Cell Reports* 3, 350-358.
- Marbach D, Costello JC, Kuffner R, Vega NM, Prill RJ, **Camacho DM**, Allison KR, Kellis M, Collins JJ, Stolovitzky G (2012). Wisdom of crowds for robust gene network inference. *Nature Methods* 9, 796-804.
- Dwyer DJ, **Camacho DM**, Callura JM, Kohanski MA, Collins JJ (2011). Antibiotic-induced bacterial cell death exhibits physiological and biochemical hallmarks of apoptosis. *Molecular Cell* 46, 561-572.
- Modi SR, **Camacho DM**, Kohanski MA, Collins JJ (2011). Functional characterization of bacterial sRNAs using a network biology approach. *Proceedings of the National Academy of Sciences* 108, 15522-15527.
- **Camacho DM**, Collins JJ (2009). Systems biology strikes gold. *Cell* 137, 24-26.
- **Camacho D**, Vera-Licona P, Laubenbacher R, Mendes P (2007). Comparison of existing reverse engineering methods by use of an in silico system. *Annals of the New York Academy of Sciences* 1115, 73-89.
- Mendes P, **Camacho D**, de la Fuente A (2005). Modelling and simulation for metabolomics data analysis. *Biochemical Society Transactions* 33, 1427-1429.
- **Camacho D**, de la Fuente A, Mendes P (2005). The origin of correlations in metabolomics data. *Metabolomics* 1, 53-63.
- Martins AM, **Camacho D**, Shuman J, Sha W, Mendes P, Shulaev V (2004). A systems biology study of two distinct growth phases of Saccharomyces cerevisiae cultures. *Current Genomics* 5, 649-663.

Patents spanning drug discovery, ML methods, and synthetic biology.

Full publication list available on Google Scholar (7,000+ citations).

Grants

Raised >$2.4M in competitive DARPA and NIH funding to support translational bioanalytics platforms at the Wyss Institute.